

Day 4: SQL Notes form Basic to Advanced

Today's Topics : SQL Keys

- Primary Key
- Foreign Key
- Unique Key
- Composite Key
- Candidate Key
- Alternate Key
- Super Key

SQL Keys:

- Keys are fundamental components in SQL that help uniquely identify records in a database table and define relationships between different tables.
- Keys are essential components in a relational database model. They are primarily used to uniquely identify rows (tuples) within a table.
- keys help establish relationships between different columns and tables within the database.

4.1 Primary Key:

A primary key is a unique identifier for each record in a table. It ensures that no two records have the same value in this column (or combination of columns).

- It is a unique key.
- It identifies a single record at a time.
- It contains only unique values, with no duplicates.
- It cannot have NULL values.
- A primary key can consist of a single column or multiple columns in a table.

Example:

```
CREATE TABLE Students (  
  StudentID INT PRIMARY KEY AUTO_INCREMENT,  
  Name VARCHAR(50),  
  Age INT  
);
```

Here, StudentID is the primary key, uniquely identifying each student.

Usage:

- Used to quickly find records in a table.
- Ensures that each record is distinct and can be uniquely

4.2 . Foreign Key

A foreign key is a column (or a set of columns) in a table that establishes a link between two tables by referencing the primary key in another table.

- Links two tables by referencing a primary key in another table.
- Ensures referential integrity between tables.
- Can have duplicate values.
- Can contain NULL values.
- Helps maintain relationships between tables.

Example:

```
CREATE TABLE Orders (  
    OrderID INT PRIMARY KEY,  
    StudentID INT,  
    FOREIGN KEY (StudentID) REFERENCES Students(StudentID) );
```

In this case, StudentID in the Orders table refers to StudentID in the Students table.

Usage:

- Establishes relationships between tables (e.g., linking orders to the students who placed them).
- Helps maintain data consistency across related tables.

4.3 Unique Key

A unique key ensures that all values in a column (or a combination of columns) are unique. Unlike a primary key, a unique key can accept a single NULL value.

- Ensures all values in the column(s) are distinct.
- Identifies a record uniquely within a table.
- Cannot contain duplicate values.
- Can include NULL values.
- Similar to a primary key but allows for one NULL value.

Example:

```
CREATE TABLE Teachers (  
    TeacherID INT UNIQUE,  
    Email VARCHAR(100) UNIQUE,  
    Name VARCHAR(50) );
```

Here, both TeacherID and Email are unique keys, ensuring no duplicate teacher IDs or email addresses.

Usage:

- Prevents duplicate values in specific columns, such as email addresses in a user table.

4.4 Composite Key

A composite key is a combination of two or more columns that together uniquely identify a row in a table.

- Consists of two or more columns to uniquely identify a record.
- Combines multiple columns to ensure uniqueness.
- Can't have duplicate combinations of column values.
- Can include NULL values if the combination maintains uniqueness.
- Used when a single column is not sufficient for uniqueness.

Example:

```
CREATE TABLE Enrollment (  
    StudentID INT,  
    CourseID INT,  
    PRIMARY KEY (StudentID, CourseID)  
);
```

Here, neither StudentID nor CourseID alone can uniquely identify a row, but together, they form a composite key.

Usage:

- Useful in many-to-many relationships where a single column cannot uniquely identify a record.

4.5 Candidate Key

A candidate key is a column, or a set of columns, that can uniquely identify each row in a table. A table can have multiple candidate keys, but only one can be the primary key.

- A column or set of columns that can uniquely identify a record.
- Each Candidate Key is a potential primary key.
- Must have unique values for every record.
- Cannot contain duplicate values.
- There can be multiple Candidate Keys in a table.

Example:

```
CREATE TABLE Library (  
    BookID INT,  
    ISBN VARCHAR(20),  
    Title VARCHAR(100),  
    PRIMARY KEY (BookID),  
    UNIQUE (ISBN)  
);
```

In this case, both BookID and ISBN are candidate keys.

Usage:

Candidate keys provide multiple options for defining a primary key, ensuring data integrity and unique identification of records.

4.6 Alternate Key

An alternate key is any candidate key that is not selected as the primary key. It's basically a unique key that wasn't chosen as the primary key.

- A Candidate Key that is not chosen as the primary key.
- Can uniquely identify records in a table.
- Must have unique values for every record.
- Cannot contain duplicate values.
- Provides an alternative method for identifying records.

Example:

```
CREATE TABLE Employees (  
    EmployeeID INT PRIMARY KEY,  
    Email VARCHAR(100) UNIQUE,  
    PhoneNumber VARCHAR(20) UNIQUE,  
    Name VARCHAR(50)  
);
```

Explanation:

- **Primary Key:** EmployeeID is the primary key. It uniquely identifies each employee.
- **Alternate Keys:** Email and PhoneNumber are alternate keys. They are also unique identifiers but are not used as the primary key.

4.7 Super Key

A super key is a set of one or more columns that can uniquely identify a record in a table. All keys (primary, candidate, etc.) are types of super keys.

- A set of one or more columns that uniquely identifies a record.
- Includes all possible columns needed to ensure uniqueness.
- Can contain extra columns beyond what's necessary for uniqueness.
- May include duplicate values in non-essential columns.
- Can include NULL values if necessary.

Example:

```
CREATE TABLE Employees (  
    EmployeeID INT,  
    Email VARCHAR(100),  
    PhoneNumber VARCHAR(20),  
    PRIMARY KEY (EmployeeID) );
```

Here, EmployeeID, (EmployeeID, Email), and (EmployeeID, PhoneNumber) are all super keys.

Usage:

Ensures that records in the table can be uniquely identified using one or more columns.

Day 5: Sql Basic Quaries

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